OPODet: Toward Open World Potential Oriented Object Detection in Remote Sensing Images

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***Abstract*— Despite recent advances in object detection, closed-set detectors with fixed training classes often overlook or misclassify unannotated objects during testing. To address this, open world object detection (OWOD) algorithms identify and label these objects as *unknown*, better aligning with real-world scenarios and human learning. However, remote sensing images, with their arbitrary object orientations and large interclass feature disparities, pose significant challenges for these algorithms. To tackle this, we propose OPODet, an Open-world Potential Oriented object Detection framework for remote sensing images. Specifically, we incorporate the oriented unknown-aware region proposal network (OUA-RPN) into traditional oriented object detection models, enabling the network to predict potential oriented objects. To address the significant interclass feature differences among potential unknown classes, we propose a multiunknown-class clustering aligning prototype (MCAP) learning method to prevent feature collapse in the feature space. In addition, to address the lack of rotation information for potential objects, we introduce a rotation potential target consistency (RPTC) algorithm to impose explicit rotation constraints for generating more accurate potential unknown proposals. Extensive experiments on DIOR-R, DOTA-v1.0, and HRSC2016 datasets demonstrate the effectiveness of our approach in detecting potential oriented objects.**

***Index Terms*— Deep learning, feature clustering, open world tasks, oriented object detection, prototype learning.**

I. Introduction

OBJECT detection aims to locate and identify all objects of interest within test images. With the rapid advancement of deep learning, neural network-based detection algorithms have become prevalent. These algorithms typically employ either convolutional [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17] or transformer-based frameworks [18], [19], [20], [21], [22], [23], [24]. However, real-world images frequently include objects from classes that are not represented in the training set, posing a significant challenge for models trained on closed-set datasets to accurately detect. These unlabeled objects may be ignored or misclassified, limiting the practical utility of detection algorithms. To address this, open world object detection (OWOD) algorithms [25], [26], [27], [28], [29] have been proposed. These algorithms not only detect labeled objects but also identify and categorize unlabeled objects as *unknown*. Open world detection algorithms better align with real-world scenarios and human learning patterns, garnering significant research attention in recent years.

The rapid advancement of technology has significantly increased satellite-based remote sensing imagery. Deep learning-based remote sensing object detection is crucial in fields such as maritime safety, disaster prevention, and military reconnaissance. However, remote sensing imagery poses unique challenges for open-world object detection algorithms. First, remote sensing images exhibit significant feature differences among potential categories, but traditional OWOD frameworks classify all unknowns as a single *unknown* class, causing feature collapse. Second, due to arbitrary orientations and dense arrangements of objects, traditional algorithms use oriented bounding boxes (OBBs) to surround targets. However, OWOD networks cannot model and represent accurate rotation information for objects from unknown classes without ground truth labels. Consequently, traditional OWOD algorithms face difficulties in accurately detecting potential oriented objects in remote sensing images.

To address these challenges, we propose a novel object detection framework named OPODet, tailored for Open world Potential Oriented object Detection in remote sensing images. This framework can predict known and unknown oriented objects simultaneously as in Fig. 1. We first introduce oriented unknown-aware region proposal network (OUA-RPN) inspired by [25] into the oriented object detector architecture. This modification enables traditional oriented object detectors to have the capability of detecting objects from unknown classes. Second, considering the significant feature differences among remote sensing object categories, we propose a multiunknown-class clustering aligning prototype (MCAP) learning method to prevent feature collapse of unknown classes in the feature space. Specifically, we adaptively cluster features of unknown classes into multiple clusters and update prototype features by aligning them with clusters. During training iteration, we utilize the prototype features to reclassify unknown-class proposals and compute a contrastive clustering loss to increase the discriminability of features. Third, to address the issues of lacking rotation information for potential objects, a rotation potential target consistency (RPTC) algorithm is proposed to explicitly impose rotation constraints on the generation of potential proposals. We apply random rotations to input images and constrain the shape and location of oriented unknown proposals before and after

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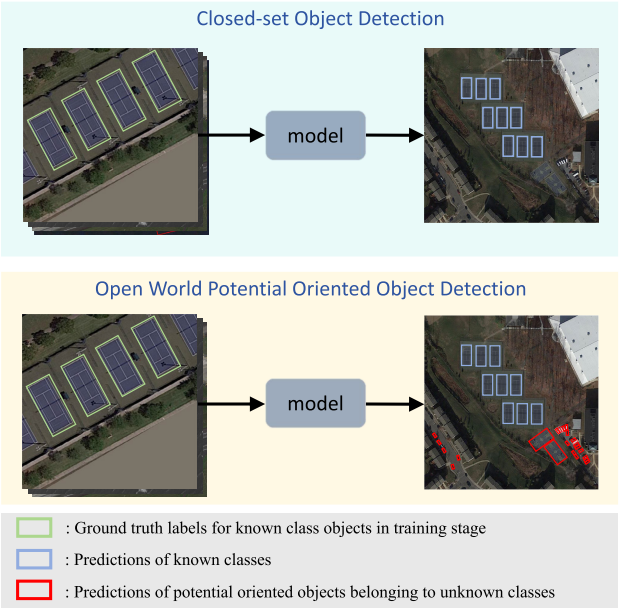


Fig. 1. In closed-set object detection, classes of the training set and the testing set are the same, and objects of other classes are simply ignored or mistakenly classified into known classes, while in open world oriented object detection, the unknown oriented objects will be detected as *unknown* using OBBs.

rotation. This helps neural networks learn consistent feature information of potential objects with different orientations.

Extensive experiments are conducted on three remote sensing image datasets: DIOR-R, DOTA-v1.0, and HRSC2016, which verify the effectiveness of our method. Our main contributions are summarized as follows.

1) A novel object detection framework named OPODet is proposed, tailored for open world potential oriented objects in remote sensing images. Our method expands the application scope of object detection in real remote sensing scenarios with abundant objects from unknown classes.
2) An MCAP learning method is proposed to prevent feature collapse of unknown classes in the feature space, thereby enhancing feature diversity and discriminability of different unknown features.
3) To address the issue of missing orientation information for potential objects, the RPTC algorithm is devised, enhancing the network's capability to learn consistent feature information of potential unknown objects with different orientations.

The rest of this article is organized as follows. Section II presents the work in the relative field. In Section III, the proposed method is introduced in detail. Section IV provides an evaluation of our method on three remote sensing image datasets, along with an analysis of its effectiveness. Finally, in Section V, we present the conclusion of this study.

II. Related Work

A. Open World Recognition and Detection

Open world task utilizes dynamic datasets to learn about unlabeled objects without prior information. The open world model can be continuously updated by integrating new known classes through continuous learning algorithms [30], [31], [32]. The pioneering work in [33] introduced an open world recognition method and established a protocol for evaluating these systems. Joseph et al. [25] introduced the OWOD problem and proposed the ORE framework to detect both known- and unknown-class proposals. The open world detection transformer (OW-DETR) in [26] enhanced the performance using multiscale self-attention and deformable receptive fields. In addition, Zhao et al. [27] extended the OWOD framework with an auxiliary proposal advisor and a class-specific expelling classifier. Dong et al. [28] developed a two-stage training approach based on Deformable-DETR [22] to mitigate catastrophic forgetting of unknown-class annotations. Yu et al. [34] proposed an efficient OWOD solution using discriminative class prototype learning. Recently, Wu et al. [29] introduced an unknown-classified OWOD (UC-OWOD) task.

OWOD algorithms align well with real-world scenarios and human learning patterns. They enable detection models to uncover unknown information and facilitate continuous learning with fewer labels available. Therefore, integrating these algorithms into remote sensing imagery is crucial. However, the arbitrary orientation of objects in remote sensing images presents a challenge for OWOD models to accurately learn and capture oriented unknown proposals without label information. This limitation significantly impedes their application in remote sensing object detection, which our method aims to overcome.

B. Few-/Zero-Shot Object Detection

Few-shot object detection (FSOD) methods enable a model to achieve high-precision detection of novel classes by fine-tuning with only a few images. FSOD methods could be divided into meta-learning-based methods and transfer learning-based methods. Typical meta-learning-based methods adopted episodic training to acquire task-level knowledge. Meta YOLO [35] and Meta R-CNN [36] used meta-weighting rather than image features. Vu et al. [37] introduced a "baby detector" as the initial detector. On the other hand, transfer learning-based approaches provide simpler and more efficient solutions. TFA [38] employed a straightforward two-stage fine-tuning strategy. SRR-FSD [39] utilized a word embedding network to expand existing prior knowledge. In addition, DMNet [40] proposed a decoupled representation transformation module to reduce the adverse impact of designed anchors. FSOD methods still require a certain amount of novel class data to train the network and need to define the novel classes in advance for class-specific detection. In contrast, our method is designed to handle any new class and can detect potential objects and classify them into "unknown" without the need for labels of novel classes.

Inspired by zero-shot learning (ZSL) [41], [42], which identifies unseen classes by transferring knowledge from seen ones, the more challenging zero-shot object detection (ZSOD) [43], [44], [45], [46], [47], [48], [49] task has garnered significant attention. ZSOD not only classifies unseen objects but also localizes them. ZSOD focuses on either embedding-based or generative-based methods. Embedding-based ZSOD [44], [45] aimed to learn a visual-to-semantic projection to align the two spaces. Generative-based methods employ generative

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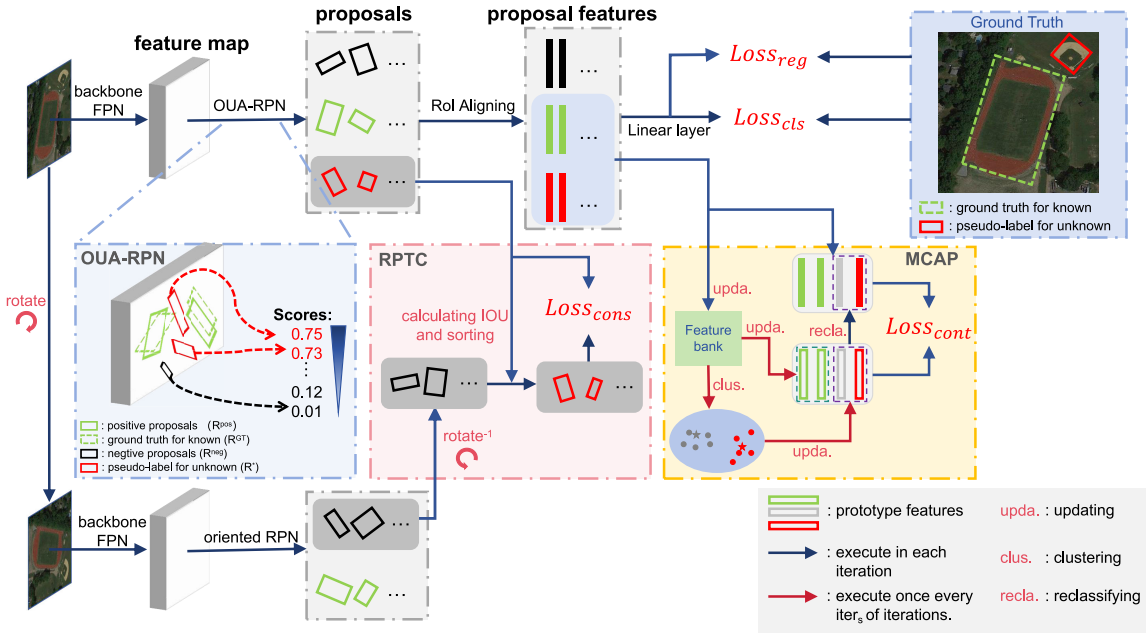


Fig. 2. Approach overview: OPODet extends upon ORCNN by replacing the oriented RPN with an OUA-RPN to detect rotated objects from unknown labels. We introduce the MCAP learning method to adaptively cluster features of unknown classes, preventing feature collapse. In addition, the RPTC algorithm imposes rotation constraints on unknown-class objects lacking angle information, thereby improving proposal accuracy.

adversarial networks (GANs) to generate visual samples of unseen classes [46], obtaining classifiers and regressors under supervised paradigms. Recent studies have introduced robust region feature synthesizers [47] and specific losses [48] to maintain interclass structure and enhance intraclass diversity. Zang et al. [49] highlighted the crucial role of category embedding representation in the ZSOD performance. ZSOD and OWOD both aim to detect objects unseen during training. However, ZSOD relies on fixed, predetermined unknown classes, requiring prior semantic knowledge or generative models for class-specific unseen samples. In contrast, OWOD methods can detect all potential objects without prior knowledge, handling entirely unknown data and distributions. This distinction enables OWOD to effectively model orientation, a challenge for ZSOD methods, making OWOD particularly suitable for remote sensing images with complex backgrounds, arbitrary object orientations, and significant semantic differences between classes.

C. Oriented Object Detection

In contrast to general object detectors [1], [2], [3], [4], [50] that use horizontal bounding boxes (HBBs), oriented object detectors employ OBBs for more precise orientation information. This is crucial for detecting objects in remote sensing images, which exhibit arbitrary orientations, large aspect ratios, and dense distributions. Advanced methods for oriented object detection [5], [6] demonstrated promising performance on remote sensing datasets. Techniques, such as [7] and [51], utilized numerous rotated anchors to enhance regression accuracy, albeit at increased computational complexity. Ding et al. [6] introduced a region of interest (RoI)-Transformer to convert horizontal RoIs into RRoIs, reducing the necessity for extensive anchor use. Other approaches, such as [8] and [9], leveraged quadrilaterals and masks for precise object delineation. Han et al. [11] proposed a feature alignment module and an oriented detection module to address alignment issues. Xie et al. [13] developed an oriented region proposal network (RPN) and R-CNN head for precise OBBs, while Han et al. [12] used rotation-equivariant networks for accurate feature extraction. Liu et al. [15] and [16] enhanced the performance of object detection algorithms on remote sensing images through the proposed adaptive anchor networks and knowledge distillation, respectively. Yao et al. [52] introduced a novel bounding box representation to handle boundary discontinuities, and Cheng et al. [53] and Gong et al. [54] employed dual-related techniques to enhance rotation detector quality. Recently, Tan et al. [14], Hua et al. [55], and Yue et al. [56] generated pseudo-labels from limited information to detect oriented objects under weakly labeled conditions. Gaussian-based [57], [58], [59] and point-set [60], [61], [62] methods have been proposed to improve the oriented object detection performance. Meanwhile, DETR-based oriented detectors [63], [64] have also made significant progress.

However, all these methods focus on detecting labeled objects in a closed set and are unable to predict potential objects without ground truth labels. In contrast, our method predicts oriented objects from both labeled known classes and unlabeled unknown classes in remote sensing images.

III. Method

A. Overview of the Proposed Method

The overall framework of our approach is illustrated in Fig. 2. Initially, input images I undergo arbitrary rotations to produce rotated images I_{rot}. These images are then simultaneously input into a backbone neural network and a feature pyramid network (FPN) [65] to generate two sets of multilevel feature maps. Subsequently, these feature maps are individually processed by an OUA-RPN to extract proposals R, along

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with their corresponding feature vectors F and objectness scores S. The proposals encompass positive samples representing instances from both known and unknown classes, as well as negative samples representing background and noise.

To address the issue of feature collapse among potential objects from unknown classes, we introduce MCAP learning to enhance feature diversity and discriminability. This approach ensures that potential objects from different categories in real scenes naturally cluster in the feature space.

In addition, we utilize the RPTC algorithm to compute the consistency loss L_{cons} for unknown proposals between images I and I_{rot}. This encourages the network to learn consistent orientation information for potential targets, thereby improving the accuracy of unknown proposals.

Simultaneously, positive and negative sample proposals extracted from the original image I, along with their corresponding feature vectors F, are input into an oriented RoI Head [13] consisting of a regression head and a class head. The regression head employs two linear layers to predict the location (x, y, w, h, θ) for each feature vector. Since objects from unknown classes lack labels, the locations of positive proposals belonging to unknown categories are treated as pseudo-labels. The regression loss L_{reg}, which measures the disparity between predictions and ground truth labels, is computed using the Smooth $L1$ loss function as defined in [66]

$$L_{\text{reg}} = \frac{1}{N}\sum_{i=1}^{N} \text{Fun}_{\text{reg}}\left(r^i, \bar{r}^i\right) \tag{1}$$

where N (default $N = 512$) denotes the total number of proposals in a mini-batch, r^i represents the location prediction from the regression head, $\bar{r}^i$ denotes the ground truth OBB of the target, and Fun_{reg} refers to the Smooth $L1$ loss function.

Similarly, the classification head utilizes two linear layers to predict the class of each feature vector, encompassing known classes and an additional class labeled as *unknown*. Positive proposals from unknown categories are treated with *unknown* as pseudo-labels. The classification loss function is defined as

$$L_{\text{cls}} = \frac{1}{N}\sum_{i=1}^{N} \text{Fun}_{\text{cls}}\left(p^i, \bar{p}^i\right) \tag{2}$$

where $\bar{p}^i$ represents the ground truth class label of the ith anchor, p^i denotes the output from the class head, and Fun_{cls} corresponds to the cross-entropy loss.

The overall loss function for the network is

$$L = \alpha_1 L_{\text{cons}} + \alpha_2 L_{\text{cont}} + \alpha_3 L_{\text{reg}} + \alpha_4 L_{\text{cls}} \tag{3}$$

where α are weighting factors assigned to each loss component. L_{cons} is the rotation-consistency loss computed in RPTC, and L_{cont} is the contrastive clustering loss in MCAP, as shown in Fig. 2.

B. Oriented Unknown-Aware RPN

To enable the detection of potential oriented objects from unknown classes, generating pseudo-labels for *unknown* during training is crucial. Joseph et al. [25] introduce an unknown-aware RPN (UA-RPN) for this purpose with horizontal objects. Similarly, OPODet uses the OUA-RPN to automatically generate pseudo-labels for oriented objects from unknown classes, as shown in Fig. 2.

Auto-Labeling Oriented Unknowns: Given an input image I, the OUA-RPN generates a set of OBB predictions R for both foreground and background instances, along with corresponding objectiveness scores S. Among these, proposals with intersection over union (IoU) ratios exceeding a certain threshold ($\text{thres}_{\text{pos}}$) with respect to ground truth labels of known classes R^{GT} are identified as positive samples R^{pos} with scores S^{pos}, while the rest are treated as negative samples R^{neg} with scores S^{neg}. Leveraging the backbone's ability to capture objects with similar semantics and strong saliency, proposals from negative samples are ranked based on scores in descending order. Then, the top k proposals from R^{neg} with the highest scores are selected as potential oriented object proposals R^* from unknown classes as

$$S^* \leftarrow \text{Top}_k(S^{\text{neg}}). \tag{4}$$

These proposals R^* are integrated into the ground truth labels as pseudo-labels $\text{GT}_{\text{unknown}} = \{\bar{x}^*, \bar{y}^*, \bar{w}^*, \bar{h}^*, \textit{unknown}\}$ to guide the network in detecting potential unknown objects.

This approach, as demonstrated in [25], has proven effective despite its simplicity. While it may introduce some noisy pseudo-labels, the network progressively learns the shapes and higher level semantics of known classes during training. This improves its capability to model potential objects and enhances prediction accuracy. However, accurately determining the orientation of unlabeled potential objects remains challenging without any ground truth information.

Upon obtaining pseudo-labels for unknown objects, we select N_k positive proposals for known classes and N_u positive proposals for unknown classes. N_k is dynamically determined based on input images and ground truth, akin to Faster RCNN [66], while N_u is a predefined hyperparameter. Their respective feature vectors are denoted as $F^{\text{pos}}_{\text{known}} = \{f^{\text{pos}}_{\text{known},i}\}_{i=1}^{N_k}$ and $F^{\text{pos}}_{\text{unknown}} = \{f^{\text{pos}}_{\text{unknown},i}\}_{i=1}^{N_u}$, with corresponding class labels c_i aligned to their ground truth labels or pseudo-labels. These positive proposal features and updated class labels are then input into the oriented RoI Head to predict outcomes and compute regression and classification losses.

C. MCAP Learning

In OUA-RPN, treating all potential objects from different classes as a single *unknown* class leads to feature collapse, which is particularly challenging in remote sensing images due to significant variations among unknown objects. To address this issue, we propose the MCAP learning method, as outlined in Fig. 3. MCAP introduces multiple prototype features for unknown classes and implements an adaptive alignment update strategy to enhance feature diversity. It consists of two main components: updating prototype features for multiple unknown classes and applying multiunknown-class contrastive clustering loss. The pseudocode for MCAP is presented in Algorithm 1.

1) Multiunknown-Class Prototype Feature Updating: Before training, a feature bank is initialized as multiple empty queues, each designated to store feature vectors for a known class or an unknown class, with a maximum length of v. During each training iteration, positive feature vectors $F^{\text{pos}} = \{F^{\text{pos}}_{\text{known}}$ and $F^{\text{pos}}_{\text{unknown}}\}$ are added to the corresponding queues

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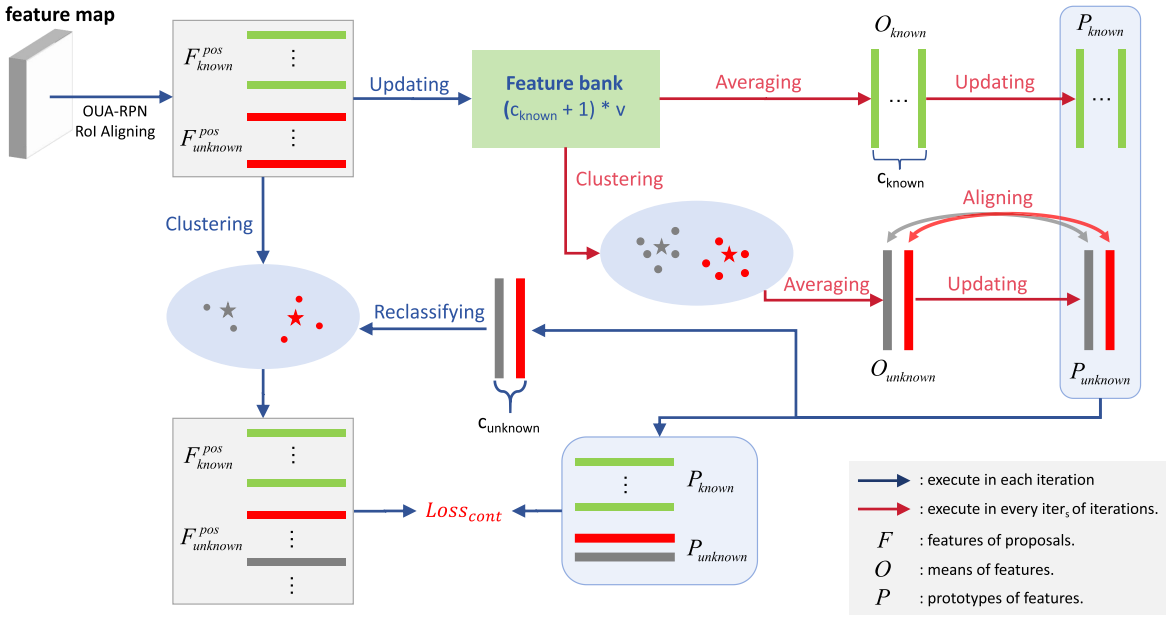


Fig. 3. MCAP procedure involves updating prototype features for multiple unknown classes every iter_s iterations using features stored in a feature bank. In each iteration, these prototype features are employed to reclassify unknown features from the OUA-RPN. Subsequently, a contrastive loss is computed between these features and the prototype features.

Algorithm 1 MCAP

Input: Positive feature vector F^{pos}, feature bank FB, update cycle for FB: $iter_s$, momentum coefficient η, cluster method $cluster()$, alignment method $align()$.
Output: Contrastive clustering loss L_{cont}.

```
Initialize P_known, P_unknown in FB.
for i in iterations:
  Update FB by adding F^pos.
  if i % iter_s == 0:
    O_known ← average(F_known) in FB
    P_known ← η P_known + (1 − η) O_known as Eq. 5
    O_unknown ← average(cluster(F_unknown)) in FB.
    O'_unknown ← align(O_unknown, P_unknown)
    P_unknown ← η P_unknown +
                (1 − η) O'_unknown as Eq. 7
  end if
  Reclassify F^pos using P_known and P_unknown.
  Calculate L_cons (Eq. 9, 10).
end for
return L_cont
```

based on their class labels. When a queue reaches its maximum length, the earliest feature is removed, and the newest feature is added to the end.

At the start of training, the number of unknown prototypes C_{unknown} is manually defined. It is important to note that C_{unknown} signifies the number of clustering clusters, not necessarily reflecting the exact number of unknown classes. A higher value indicates a wider distribution of features for unknown classes in the feature space. Prototype feature vectors $P_{\text{known}} = \{p_{\text{known},i}\}_{i=1}^{C_{\text{known}}}$ and $P_{\text{unknown}} = \{p_{\text{unknown},i}\}_{i=1}^{C_{\text{unknown}}}$ are initialized for C_{known} known classes and C_{unknown} unknown classes, respectively. During training, after a specified number of iterations iter_s, prototype feature vectors are updated as follows.

For all known-class features in the feature bank, compute the mean feature vector $O_{\text{known}} = \{o_{\text{known},i}\}_{i=1}^{C_{\text{known}}}$ specific to each class. Then, update the prototype feature vectors P_{known} using a momentum coefficient η

$$P_{\text{known}} = \eta P_{\text{known}} + (1-\eta) O_{\text{known}}. \tag{5}$$

Every iter_s iteration, for unknown-class features in the feature bank, retrieves all unknown-class feature vectors from the current iteration. Perform K-means clustering with C_{unknown} clusters on these vectors to compute the feature vector means $O_{\text{unknown}} = \{o_{\text{unknown},j}\}_{j=1}^{C_{\text{unknown}}}$ for each cluster. To align O_{unknown} with the unknown-class prototype features, calculate the distance between each prototype feature vector $p_{\text{unknown},i}$ and the mean feature vector $o_{\text{unknown},j}$

$$D(p_i, o_j) = \| p_{\text{unknown},i} - o_{\text{unknown},j} \|_p \tag{6}$$

where p is the norm order, set to 10 in our experiments. Apply the Hungarian matching algorithm as in [18] to match these vectors, and reorder $o_{\text{unknown},j}$ accordingly to align the sets, resulting in the new set $O'_{\text{unknown}} = \{o'_{\text{unknown},j}\}_{j=1}^{C_{\text{unknown}}}$. Then, update P_{unknown} using O'_{unknown}

$$P_{\text{unknown}} = \eta P_{\text{unknown}} + (1-\eta) O'_{\text{unknown}}. \tag{7}$$

By continuously optimizing prototype features across multiple unknown classes during training, this method ensures that each prototype feature evolves with the network parameters, leading to more representative prototype features.

2) Multiunknown-Class Contrastive Clustering Loss: To deal with feature collapse, we first utilize multiple prototype features P_{unknown} to reclassify $F^{\text{pos}}_{\text{unknown}}$ in the current iteration by incorporating them into the most similar clusters of

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unknown classes

$$\mathrm{Cls}\left(f^{\mathrm{pos}}_{\mathrm{unknown},j}\right) = C_{\mathrm{known}} + \arg\max_{j} \frac{f^{\mathrm{pos}}_{\mathrm{unknown},i} \cdot p_{\mathrm{unknown},j}}{\|f^{\mathrm{pos}}_{\mathrm{unknown},i}\| \cdot \|p_{\mathrm{unknown},j}\|} \tag{8}$$

where $\mathrm{Cls}(f^{\mathrm{pos}}_{\mathrm{unknonw},j}) \in [C_{\mathrm{known}}+1, C_{\mathrm{known}}+C_{\mathrm{unknown}}]$.

For each positive feature vector f_c, where c denotes its new corresponding class, the contrastive loss can be computed using the following formula:

$$L_{\mathrm{cont}}(f_c) = \sum_{i=1}^{C_{\mathrm{known}}+C_{\mathrm{unknown}}} l(f_c, p_i), \quad \text{where}$$
$$l(f_c, p_i) = \begin{cases} D(f_c, p_i), & \text{if } i = c \\ \max\{0, \delta - D(f_c, p_i)\}, & \text{otherwise} \end{cases} \tag{9}$$

where the function $D(.,.)$ calculates the p-norm distance between two vectors, and its computation is the same as in (6). δ is a hyperparameter that ensures the nonnegative property and defines the maximum value of this part of loss when $i \neq c$.

The overall contrastive clustering loss is given by

$$L_{\mathrm{cont}} = \frac{1}{N_k + N_u} \sum_{q=1}^{N_k+N_u} L_{\mathrm{cont}}(f_{c,q}) \tag{10}$$

where q represents the qth positive proposal feature vector.

Through the computation of the aforementioned contrastive clustering loss for multiple unknown classes, various unknown clusters will adaptively form in the feature space, thereby preserving the interclass feature diversity and avoiding feature space collapsing.

D. RPTC Algorithm

Without ground truth labels for potential objects from unknown classes, OUA-RPN faces challenges in obtaining accurate oriented pseudo-labels. To address this, we propose the RPTC algorithm detailed in Algorithm 2. RPTC employs a self-supervised learning approach that imposes consistency constraints on unknown proposals, facilitating the learning of consistent feature representations across various rotation angles.

Initially, each input image I undergoes random rotations by angles 90°, 180°, and 270° to produce rotated images I_{rot}

$$I_{\mathrm{rot}} = \mathrm{Rot}(I). \tag{11}$$

Both I and I_{rot} are then fed into the backbone, FPN, and OUA-RPN in tandem, resulting in proposal sets $R = \{r^i\}$ and $R_{\mathrm{rot}} = \{r^i_{\mathrm{rot}}\}$ with corresponding confidence scores $S = \{s^i\}$ and $S_{\mathrm{rot}} = \{s^i_{\mathrm{rot}}\}$. Proposals belonging to unknown classes are denoted as $R^* = \{r^{*i}\}_{i=1}^k$.

Next, R_{rot} is aligned with R based on their rotation angles to obtain $R_{\mathrm{rot_new}}$, which are then projected into the same coordinate space

$$R_{\mathrm{rot_new}} = \mathrm{Rot}^{-1}(R_{\mathrm{rot}}). \tag{12}$$

For each r^{*i} in R^*, the proposal in $R_{\mathrm{rot_new}}$ with the highest IoU is identified as $r^{*i}_{\mathrm{rot_new}}$, forming a pair of rotation-consistent proposals. Finally, the generalized IoU (GIoU) [67] between each pair of rotation-consistent candidate boxes is computed as the rotation-consistency loss to supervise the network

$$L_{\mathrm{cons}} = -\frac{1}{k} \sum_{i=1}^{k} F_{\mathrm{ori}}\left(r^{*i}, r^{*i}_{\mathrm{rot_new}}\right) \tag{13}$$

where $F_{\mathrm{ori}}(.,.)$ denotes the GIoU loss function for rotated boxes, similar to [68]. These operations enable the neural network to extract directional feature information for potential classes, thereby improving the quality of generated pseudo-labels for potential objects significantly.

Algorithm 2 RPTC

Input: Input image I, backbone, FPN, and OUA-RPN $G()$, ground truth of known classes $\{r^*, p^*\}$.
Output: Rotation-consistency loss L_{cons}.

```
Rotate I randomly: I_rot = Rot(I)
R, S ← G(I, {r*, p*}); R_rot, S_rot ← G(I_rot, {r*, p*})
Obtain R^pos and S^pos as in [66]
S^neg ← S/S^pos
Obtain R* by finding S* ← Top_k(S^neg)
R_rot_new = Rot^{-1}(R_rot)
for each r*_j in R*:
    r*_{rot_new,j} = arg max_{r_i ∈ R_rot_new} IoU(r_i, r*_j)
end for
Calculate L_cons as in Eq. 13
return L_cons
```

IV. Experiments

A. Datasets and Evaluation Metrics

Comprehensive experiments were conducted on three challenging datasets: DIOR-R [69], DOTA-v1.0 [70], and HRSC2016 [71], to evaluate the proposed method.

1) *DIOR-R* [69] comprises high-resolution images from aerial and ground perspectives, featuring diverse object types such as vehicles, people, and buildings. The dataset includes 23 463 images, each sized 800 × 800 pixels, with 192 472 instances covering 20 object classes: airplane (AP), airport (AT), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), dam (DA), expressway service area (ES), expressway toll station (ET), harbor (HB), golf field (GF), ground track field (GTF), overpass (OP), ship (SH), stadium (SD), storage tank (ST), tennis court (TC), train station (TS), vehicle (VH), and windmill (WM).
2) *DOTA-v1.0* [70] consists of 2806 large remote sensing images ranging from 800 × 800 to 4000 × 4000 pixels, containing 188 282 instances across 15 categories: plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC).
3) *HRSC2016* [71] presents a significant challenge in ship detection with OBB annotations across 1061 aerial images ranging from 300 × 300 to 1500 × 900 pixels. It is divided into training, validation, and test sets with 436, 181, and 444 images, respectively. Both the training

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TABLE I
COMPOSITION OF DATASETS FOR KNOWN AND UNKNOWN CLASSES

Dataset	Setting	Classes split	Train instances	Test instances
DIOR-R	16 + 4	AP, BF, BR, CH, ES, ET, DA, GF, HB, OP, SH, SD, ST, TC, TS, VH	68073	116750
		AT, BC, GTF, WM	0	7695
DOTA-v1.0	11 + 4	LV, BR, PL, SH, BC, GTF, SV, BD, ST, RA HA	93931	25680
		HC, TC, SBF, SP	0	1264
DOTA-v1.0	8 + 7	LV, SP, HC, BR, PL, SH, SBF, BC	33961	16677
		GTF, SV, BD, TC, ST, RA, HA	0	10267
HRSC	15 + 4	ship, air, war., mer., Nim., Ent., Arl., Whi., Per., San., Tic., Aus., Tar., Con., Com.A	1377	917
		Car.A, Con.A, Med., Car.B	0	161

and validation subsets are used for model training, while the test set is reserved for evaluation. The dataset utilizes 19 class labels as per [72].

To address the open world potential oriented object task, we adopt a known + unknown setting. During training, labels for unknown classes are omitted. In the testing phase, all instances from unknown classes are treated as unknown. Table I details the dataset splits, with DIOR-R adopting a 16 + 4 setting and DOTA-v1.0 using 11 + 4 setting for comparison, as in [49]. To further evaluate the generalization and effectiveness of our methods, we have introduced an additional 8 + 7 setting for DOTA-v1.0 and a 15 + 4 setting for HRSC2016.

To assess our method's performance, we use average precision (AP) to measure average detection accuracy via the precision–recall curve area. For evaluating open world potential object detection, we denote known mean AP (k-mAP) as known-class objects' average mean AP (mAP) and unknown AP (uAP) as unknown-class objects' AP. Furthermore, we employ the wilderness impact (WI) metric as in [25] to quantify misidentified unknown-class targets as known-class ones. TP_k and FP_k represent true positive and false positive proposals for known classes, respectively, while FP_o indicates unknown objects mistakenly classified as known classes. Unknown recall (uR) is also employed to measure the algorithm's recall rate for unknown-class targets

$$\mathrm{WI} = \frac{\mathrm{FP}_o}{\mathrm{TP}_k + \mathrm{FP}_k}. \tag{14}$$

B. Implementation Details

Our methodology builds upon the foundational architecture of ORCNN [13], adopting similar hyperparameter settings for backbone, FPN, RPN, and RoI Head. Specifically, we utilize ResNet-50 [73] as the backbone, an oriented RPN Head for the RPN module, and an OBB standard RoI Head for the RoI module, consistent with the specifications outlined in [13]. The model is optimized using stochastic gradient descent (SGD) with a learning rate of 0.005 and a momentum coefficient of 0.9. Training spans 12 epochs for DIOR-R and DOTA-v1.0 datasets, and 36 epochs for HRSC2016. During training, N_u is set to 10 for DIOR-R and DOTA-v1.0, and 5 for HRSC2016. We maintain three unknown clusters and update prototype features every 50 iterations ($\mathrm{iter}_s = 50$) with a momentum coefficient (η) of 0.99. The loss weights (α_1–α_4) are set to (0.01, 0.05, and 1, 1), respectively. Input images are randomly rotated by angles from {90°, 180°, 270°} to augment training data. The max length of feature bank v is set to 50.

TABLE II
DETECTION RESULTS (%) OF DIFFERENT METHODS UNDER 16 + 4 SETTING ON THE DIOR-R TEST SET (%). (↑) INDICATES THAT HIGHER VALUES OF THE METRIC ARE DESIRABLE, WHILE (↓) SUGGESTS THAT LOWER VALUES ARE PREFERABLE

Methods	Format	k-mAP (↑)	uAP (↑)	uR (↑)	WI (↓)
RetinaNet [2]	OBB	57.0	0	0	0
Faster RCNN [66]	OBB	62.1	0	0	0
S^2ANet [11]	OBB	62.7	0	0	0
ORCNN [13]	OBB	62.5	0	0	0
ReDet [12]	OBB	68.4	0	0	0
LSKNet [74]	OBB	64.7	0	0	0
STD-O [75]	OBB	65.6	0	0	0
RRFS [47]	HBB	41.9	2.8	19.9	-
ContrastZSD [76]	HBB	51.4	3.9	25.9	-
DescReg [49]	HBB	**68.7**	7.9	34.3	-
u-ORCNN	OBB	62.3	29.3	65.5	0.76
OPODet	OBB	61.7	**32.9**	**66.3**	**0.68**

Our experiments are conducted using the PyTorch deep learning framework, implemented with Python and C++. All experiments are performed on an NVIDIA GeForce RTX 2080Ti GPU and Intel[1] Xeon[1] CPU E5-2630 v4 (2.20 GHz).

C. Results on Remote Sensing Image Dataset

In this section, the detection performance for each known class and potential unknown class is reported on DIOR-R, DOTA-v1.0, and HRSC2016 datasets as shown next. The detection performance of unknown-class objects is demonstrated through uAP, uR, and WI.

1) DIOR-R: Table II compares the results of our method with other related methods on the DIOR-R dataset. RetinaNet [2], Faster RCNN [66], S^2ANet [11], ReDet [12], LSKNet [74], and STD-O [75] are fully supervised oriented object detection methods that can detect known classes using OBBs but cannot detect unknown classes. RRFS [47], ContrastZSD [76], and DescReg [49] are ZSOD methods that can detect unknown classes. However, due to the need for additional semantic information for precise classification, their uAP values are relatively low, making them less practical for real-world tasks. In addition, due to the lack of angular perception, ZSOD methods can only detect horizontal objects using simpler HBBs, which also results in higher k-mAP. Note that u-ORCNN is an augmentation of the ORCNN [13], incorporating OUA-RPN to endow it with the capability to detect potential objects. Our method improves u-ORCNN by implementing MCAP and RPTC, which can detect potential oriented objects without significantly compromising the detection accuracy for known classes (61.7% versus 62.5%).

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TABLE III
DETECTION RESULTS (%) OF DIFFERENT METHODS ON DOTA-V1.0 TEST SET UNDER 11 + 4 SETTING

Methods	Format	k-mAP (↑)	uAP (↑)	uR (↑)	WI (↓)
RetinaNet [2]	OBB	58.2	0	0	0
Faster RCNN [66]	OBB	72.8	0	0	0
S^2ANet [11]	OBB	76.4	0	0	0
ORCNN [13]	OBB	77.1	0	0	0
ReDet [12]	OBB	**82.3**	0	0	0
LSKNet [74]	OBB	80.0	0	0	0
STD-O [75]	OBB	77.4	0	0	0
RRFS [47]	HBB	47.1	2.2	14.2	-
ContrastZSD [76]	HBB	41.6	2.8	12.2	-
DescReg [49]	HBB	68.7	4.7	29.9	-
u-Faster RCNN	OBB	70.8	33.8	72.9	0.22
OPODet (Faster RCNN)	OBB	70.8	**36.9**	74.5	0.20
u-ORCNN	OBB	76.7	33.3	78.6	0.17
OPODet (ORCNN)	OBB	76.3	36.8	**80.9**	**0.14**

TABLE IV
DETECTION RESULTS (%) OF DIFFERENT METHODS ON DOTA-V1.0 TEST SET UNDER 8 + 7 SETTING

Methods	Format	k-mAP (↑)	uAP (↑)	uR (↑)	WI (↓)
RetinaNet	OBB	63.8	0	0	0
Faster RCNN	OBB	68.9	0	0	0
S^2ANet	OBB	67.8	0	0	0
ORCNN	OBB	71.4	0	0	0
ReDet	OBB	72.7	0	0	0
LSKNet	OBB	**73.1**	0	0	0
STD-O	OBB	69.2	0	0	0
RRFS	HBB	49.2	4.6	14.1	-
M-RRFS [77]	HBB	50.2	8.8	15.2	-
u-ORCNN	OBB	71.9	19.7	60.2	3.23
OPODet	OBB	71.7	**26.7**	**61.0**	**3.19**

TABLE V
DETECTION RESULTS (%) OF DIFFERENT METHODS ON THE HRSC2016 TEST SET UNDER 15 + 4 SETTING

Methods	Format	k-mAP (↑)	uAP (↑)	uR (↑)	WI (↓)
RetinaNet	OBB	33.4	0	0	0
Faster RCNN	OBB	56.7	0	0	0
S^2ANet	OBB	65.5	0	0	0
ORCNN	OBB	71.5	0	0	0
ReDet	OBB	**73.5**	0	0	0
LSKNet	OBB	71.6	0	0	0
STD-O	OBB	70.4	0	0	0
RRFS	HBB	69.5	1.6	11.3	-
M-RRFS	HBB	69.7	2.8	18.2	-
u-ORCNN	OBB	71.8	7.0	53.4	10.5
OPODet	OBB	70.1	**13.5**	**55.9**	**2.7**

Moreover, it achieves much higher accuracy for unknown classes compared to zero-shot methods (32.9% versus 7.9% on uAP and 66.3% versus 34.3% on uR). These results demonstrate the advantages and uniqueness of our method over fully supervised and ZSOD methods, highlighting its better application prospects in the remote sensing field.

2) DOTA-v1.0: In this section, we contrast our method with other oriented object detection approaches on DOTA-V1.0 using 11 + 4 and 8 + 7 settings.

Table III presents the results of our method on the DOTA-v1.0 dataset using the 11 + 4 setting. As observed from the table, our approach not only effectively detects oriented objects of known classes but also demonstrates strong detection performance for potential oriented objects of unknown classes. Compared to ZSOD methods that require more prior knowledge to finely classify unknown objects, our method significantly outperforms them in terms of unknown object detection. Although our method cannot perform fine-grained classification of unknown classes, its detection performance and unique capability to detect rotated objects in remote sensing images make it an indispensable approach. In addition, in this set of experiments, we integrated our proposed method into the Faster R-CNN framework. The experimental results clearly demonstrate that our method consistently enhances the model's ability to detect unknown object classes across different frameworks.

Table IV provides detailed results of the detection performance of our method for each known and unknown class under the 8 + 7 setting. The results demonstrate that even with a limited number of classes designated as known, our method is capable of detecting potential objects effectively. Moreover, our method significantly outperforms ZSOD methods RRFS [47] and M-RRFS [77] in terms of uAP and uR by detecting OBBs, demonstrating the unique advantages of our approach.

3) HRSC2016: HRSC2016 is a specialized dataset for ships, where the interclass differences between different categories are minimal (all belonging to the *ship* class). Table V presents the performance metrics on the HRSC2016 test set after adopting the 15 + 4 setting. From the table results, although our method slightly reduces k-mAP, there is a significant improvement in the detection performance of potential objects from unknown categories (uAP increased from 7.0% to 13.5%, uR increased from 53.4% to 55.9%, and WI decreased from 10.5% to 2.7%). Similar to Table IV, our method's detection performance for unknown objects is significantly higher than that of existing zero-shot detection methods.

D. Ablation Studies

To better understand how the proposed method works, we conduct a series of ablation studies on the DOTA-v1.0 dataset using the 8 + 7 setting and HRSC2016 using the 15 + 4 setting.

1) Effectiveness of Each Component: The contributions of different components of the proposed method are listed in Table VI. We adopt ORCNN as the baseline and initialize hyperparameters as in ORCNN [13] and ORE [25]. ORCNN solely detects known-class oriented objects in remote sensing images due to its lack of a component generating unknown-class proposals. Upon replacing RPN with the OUA-RPN, the model acquires the capability to detect potential unknown objects. The integration of MCAP effectively suppresses the spatial collapse phenomenon of unknown-class features, leading to a significant enhancement of uAP. Incorporation of RPTC improves the quality of oriented unknown proposals, resulting in corresponding improvements in WI, uR, and uAP compared to the versions without it. Concurrent usage of both components further elevates the detection accuracy for unknown-class targets. Although there is a slight decrease in k-mAP for known classes, likely due to the expansion of the unknown space in feature space, the metric uAP shows significant improvement with our method. This directly reflects the method's effectiveness in detecting potential objects. In addition, our method shows improvements in metrics such as WI and uR. These results demonstrate the effectiveness of each

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TABLE VI
EFFECTIVENESS OF EACH COMPONENT OF THE PROPOSED METHOD IN THE DOTA-V1.0 DATASET (%)

Dataset	Component	Baseline	Different Settings			
	OUA-RPN		✓	✓	✓	✓
	MCAP			✓		✓
	RPTC				✓	✓
DOTA-v1.0	k-mAP (↑)	71.37	71.93	71.59	**71.99**	71.68
	WI (↓)	-	3.23	3.19	**3.18**	3.19
	uR (↑)	-	60.23	60.33	60.64	**60.96**
	uAP (↑)	-	19.70	24.00	22.87	**26.66**
HRSC2016	k-mAP (↑)	71.51	**71.79**	70.62	70.06	70.13
	WI (↓)	-	10.5	3.24	5.51	**2.72**
	uR (↑)	-	53.42	55.21	54.95	**55.90**
	uAP (↑)	-	6.96	11.56	9.88	**13.45**

TABLE VII
RESULTS UNDER DIFFERENT HYPERPARAMETERS IN OUA-RPN (%)

N_u	5	10	20
uR (↑)	50.54	60.96	**66.19**
u-dets(↓)	**22626**	34096	48836
WI (↓)	3.41	**3.19**	3.32
uAP(↑)	24.86	**26.66**	26.54
$thre_{pos}$	0.3	0.5	0.7
uR	57.65	**60.96**	58.24
uAP	24.28	**26.66**	23.13

component for open world potential oriented object detection task.

Next, we perform the detailed ablation experiments for each proposed component on the DOTA-v1.0 dataset with the 8 + 7 setting, in order to analyze the corresponding hyperparameters.

2) Ablation Studies on OUA-RPN: The OUA-RPN generates positive sample proposals with an IoU greater than thre_{pos} for any ground truth box of known classes and produces N_u proposals for unknown classes. In [25], thre_{pos} is set to 0.5 and N_u is set to 5. We optimized thre_{pos} around 0.5 and increased N_u to account for the smaller, more numerous, and denser objects in remote sensing images. Table VII shows the results for various N_u and thre_{pos} settings, where *u-dets* indicates the number of predicted unknown objects. As N_u increases, uR rises, but with more false positives. uAP and WI are optimal when $N_u = 10$. Thus, we set $N_u = 10$ as the default for DOTA-v1.0. The best prediction performance for unknown classes occurs with $\text{thre}_{\text{pos}} = 0.5$.

3) Ablation Studies on MCAP: The experimental results under different hyperparameters in MCAP without RPTC interference are illustrated in Table VIII. In [25], the unknown class C_{unknown} is equal to 1, we increased it to enrich the unknown class in the feature space to some extent. Considering that an excessively large C_{unknown} could interfere with the feature space of known classes and increase computational costs, we chose not to further expand the size of it. It is demonstrated that expanding the unknown class from a single class to multiple classes effectively enhances the model's detection performance. Moreover, the magnitude of α_2 reflects the extent of its impact on the overall model. Based on the experiments, it is observed that the performance is relatively optimal when $\alpha_2 = 0.05$. Simultaneously, when the parameters η, δ, iter_s, and v are configured to 0.9, 2, 50, and 50, respectively, the MCAP module demonstrates a comparatively greater positive impact on the model.

TABLE VIII
RESULTS UNDER DIFFERENT HYPERPARAMETERS IN MCAP (WITHOUT RPTC) (%)

$C_{unknown}$	0	1	3	5
uR	60.23	60.25	60.33	**60.64**
uAP	19.70	21.76	**24.00**	23.54
α_2	0	0.01	0.05	0.1
uR	60.23	60.15	**60.33**	60.27
uAP	19.70	21.84	**24.00**	23.16
η	0.1	0.5	0.9	0.99
uR	60.09	60.14	**60.33**	60.31
uAP	20.26	22.45	**24.00**	23.53
δ	0.1	1	2	5
uR	59.89	60.19	**60.33**	58.46
uAP	22.13	23.65	**24.00**	21.71
$iter_s$	10	50	100	200
uR	59.69	**60.33**	60.16	59.57
uAP	23.23	**24.00**	23.56	23.44
v	10	20	50	100
uR	58.65	59.77	**60.33**	59.46
uAP	22.56	23.40	**24.00**	**24.00**

TABLE IX
RESULTS UNDER DIFFERENT HYPERPARAMETERS IN RPTC (WITHOUT MCAP) (%)

	Rotation strategy				α_1 (randomly rotated)		
	random	90°	180°	270°	0.005	0.01	0.1
uR	**60.64**	59.65	60.01	59.79	60.58	**60.64**	60.26
uAP	**22.87**	20.82	21.03	21.26	21.63	**22.87**	21.54

4) Ablation Studies on RPTC: For the RPTC part, we conducted experiments concerning the image rotation method and the weighting of the loss in this section, with the results presented in Table IX. It is evident from the table that employing random rotation enables the network to better represent the rotational information of unknown-class targets, resulting in improved detection performance. Moreover, the algorithm achieves the optimal performance when $\alpha_1 = 0.01$.

E. Qualitative Results

Fig. 4 shows the detection results on the DOTA-v1.0 dataset under the 8 + 7 setting, highlighting our method's ability to detect potential oriented objects in remote sensing images. Known-class objects are marked with green OBBs, while potential unknown-class objects are marked with red. In the first row, our method detects known-class objects effectively, even with potential object interference. The second row demonstrates our method's ability to detect semantically rich potential objects. Our approach not only accurately detects known-class objects but also predicts OBBs for potential unknown-class objects without any corresponding labels, enhancing the practicality and generality of remote sensing object detection. While the third row includes some inaccurate and false positive predictions for unknown classes, it is clear that our model excels at localizing semantically rich background regions, which is advantageous for reconnaissance, remote sensing warning, and related applications.

Fig. 5 shows the detection results of the traditional oriented object detector (ORCNN) and our method OPODet on the HRSC2016 dataset under the 15 + 4 setting and DIOR-R under the 16 + 4 setting. The first row shows that the traditional detector fails to detect potential objects, while

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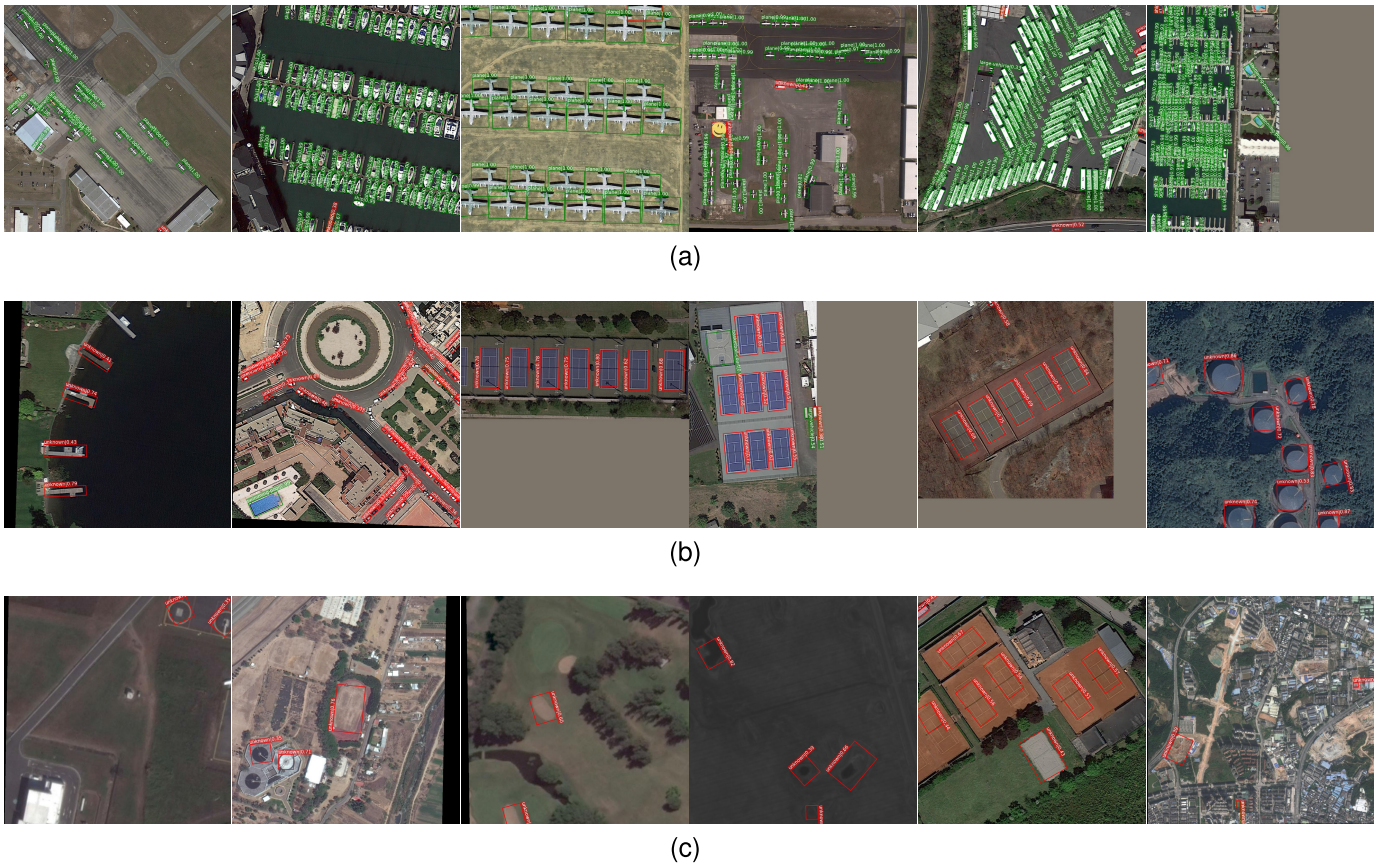


Fig. 4. (a) Detection results for known classes with unknown object interference on DOTA-v1.0 under 8 + 7 setting. (b) Detection results for potential objects from unknown classes on DOTA-v1.0 under 8 + 7 setting. (c) Images with some inaccurate unknown detection results.

our method successfully detects them with OBB. The third row presents the results obtained on the DIOR-R dataset using the 16 + 4 setting. Our method accurately detects dense oriented objects of known classes and effectively responds to unknown classes such as airports, basketball courts, ground track fields, and windmills. These results highlight the generalization and practical applicability of our approach.

F. Discussion and Analysis

1) Effectiveness of MCAP: Fig. 6 illustrates the t-distributed stochastic neighbor embedding (t-SNE) visualization of feature vectors stored in the feature bank at the end of training under the 8 + 7 setting on DOTA-v1.0. The left subfigure displays the results without alignment through multiunknown-class clustering. Despite the existence of seven unknown classes, the feature vectors collapse into a single cluster with some unknown-class features that closely resemble the bridge features. These anomalous results indicate that having only one unknown class can cause a collapse in the unknown feature space, making it difficult to encompass all potential categories and leading to misclassification.

In Fig. 6(b), the results incorporating MCAP show improved clustering of unknown-class feature vectors into distinct groups within the feature space, for example, elongated objects such as harbor (HA) cluster near bridge (BR), and rectangular sports fields such as tennis court (TC) cluster near basketball court (BC). Objects that remain distant from all classes likely represent circular objects, such as storage tank (ST), which are semantically distant from known classes. These findings demonstrate our method's ability to naturally cluster unknown objects, enhancing their distinctiveness and diversity. In addition, the approach effectively captures the objects that resemble the typical shapes and semantics of known classes (HA and TC) and emphasizes those with significant background saliency (ST).

TABLE X
MAP ON NOVEL CLASSES UNDER DIFFERENT FSOD SETTINGS (%)

Method	2-shot	5-shot	10-shot
ORCNN	26.00	46.68	57.52
u-ORCNN	29.09 (+3.09)	47.76 (+1.08)	58.55 (+1.03)
Ours	**30.73 (+4.73)**	**50.32 (+3.64)**	**59.86 (+2.34)**

2) Enhancing Performance on FSOD: Modeling potential objects from unknown classes during training enhances the model's ability to represent open world scenarios, thereby improving the detection accuracy of novel classes in subsequent FSOD tasks. By treating unknown classes as novel classes, fine-tuning the network's oriented RoI Head across different FSOD settings yields mAP scores for novel classes, as shown in Table X. Here, "1-shot" denotes training with only one randomly selected image per class. Results in the table demonstrate that enhancing the detection capability for potential objects allows the network to quickly adapt to novel class samples, resulting in superior performance in FSOD tasks. Moreover, our method outperforms u-ORCNN, highlighting its superior feature representation for potential objects.

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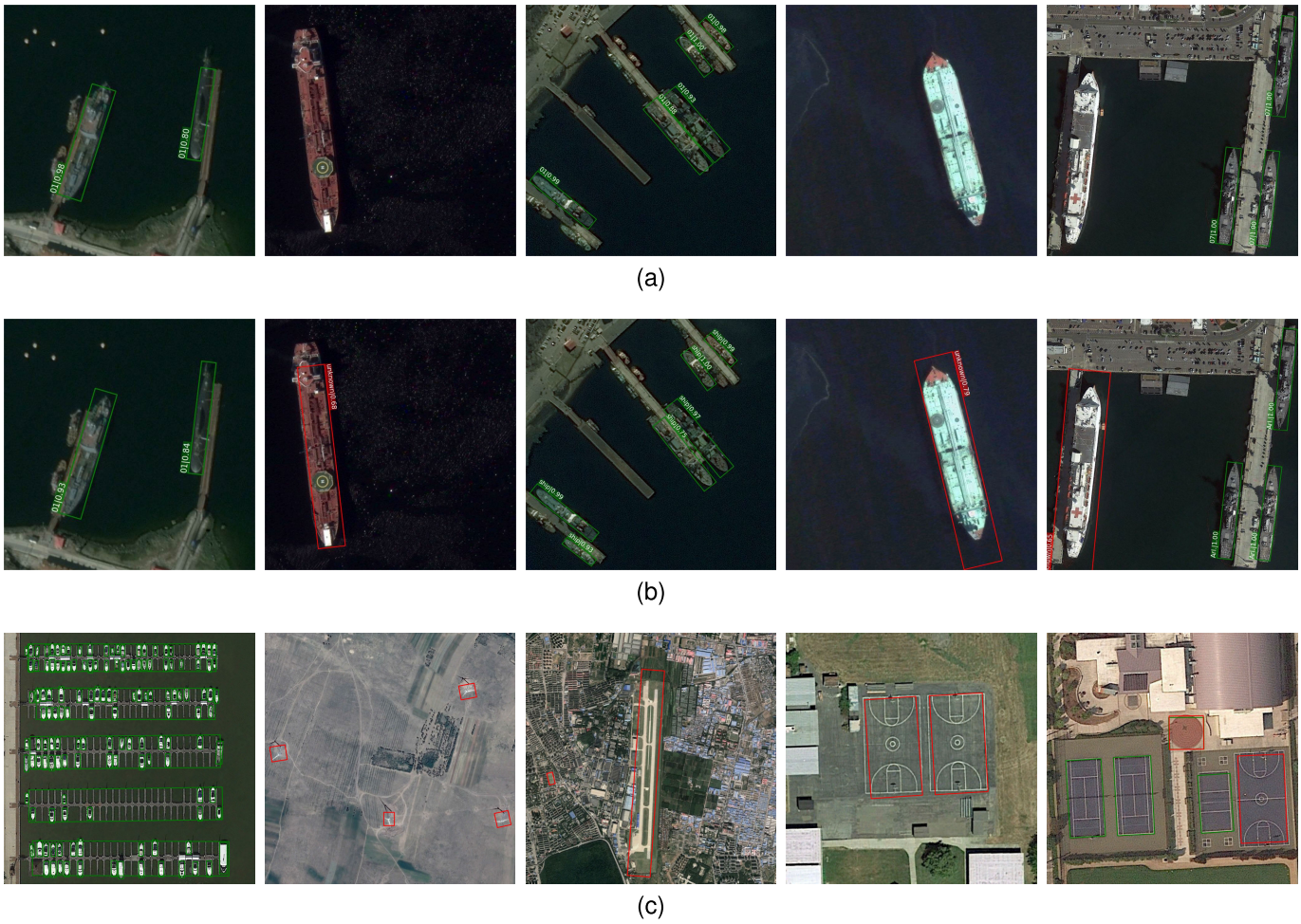


Fig. 5. (a) Detection results of ORCNN on HRSC2016 under 11 + 4 setting. (b) Detection results of OPODet (Ours) on HRSC2016 under 11 + 4 setting. (c) Detection results of OPODet on DIOR-R under the 16 + 4 setting.

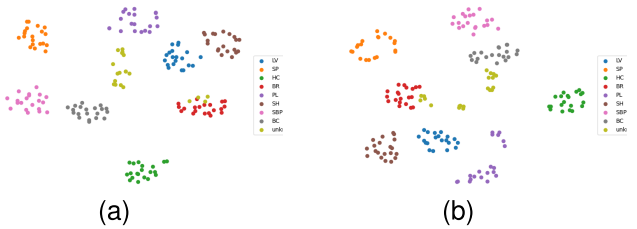


Fig. 6. (a) t-SNE visualization with one unknown cluster. (b) t-SNE visualization with multiple unknown clusters ($C_{\text{unknown}} = 3$).

TABLE XI

uAP RESULTS USING DIFFERENT UNKNOWN-CLASS SETTINGS ON THE DOTA-V1.0 DATASET (%)

unknown categories	HC, TC, SBF, SP	BD, TC, ST, RA
u-ORCNN	29.3	22.7
Ours	**32.9 (+3.6)**	**26.5 (+3.8)**

3) Results on Various Unknown-Class Settings: To further validate the impact of unknown category selection on experimental results, we compared two 11 + 4 settings using different unknown categories on the DOTA-v1.0 dataset, as shown in Table XI. The results indicate that changing the specific unknown categories affects the uAP outcome. However, our method consistently achieves significant improvements under both settings, demonstrating its robustness and generalizability.

V. CONCLUSION

In order to enhance the generalizability and practicality of object detection methods in open world remote sensing scenarios, this article proposes a method for detecting potential oriented objects in an open world setting. By introducing the OUA-RPN, MCAP, and RPTC modules and algorithms, the detection model is endowed with the capability to detect oriented potential objects while mitigating to some extent the issues of feature collapse of unknown objects and inaccurate angle prediction for unknown-class proposals. Experiments on DIOR-R, DOTA-v1.0, and HRSC2016 datasets validate the effectiveness and irreplaceability of our approach compared with ZSOD and its potential in improving the performance of FSOD. Our future research will focus on enhancing feature representation and prediction for dense small objects within unknown classes.

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